Short Notes: Peptides & Polypeptides

Definitions:

Peptides are small polymers of amino acids formed by condensation reactions.

- *Oligopeptide: A few amino acids (e.g., Dipeptide = 2, Tripeptide = 3, Decapeptide = 10).*
- *Polypeptide: Larger peptides, typically containing more than 50 amino acids (often synonymous with proteins).*

Their structure & function depend on the nature, sequence, and spatial relationship of the amino acids.

1. Peptide Bond Formation

*Amino acids link via a condensation reaction (dehydration synthesis) between the carboxyl group (-**COOH**) of one amino acid and the amino group (-**NH_2**) of another, eliminating a water (**H_2O**) molecule.*

Representation of Peptide Bond Formation:

Amino Acid 1 + **Amino Acid 2** →

$R_1\text{-}CH(NH_2)\text{-}COOH$ + $H_2N\text{-}CH(R_2)\text{-}COOH$ →

Dipeptide + H_2O

$R_1\text{-}CH(NH_2)\text{-}CO\text{-}NH\text{-}CH(R_2)\text{-}COOH$

*The **-CO-NH-** is the Amide / Peptide Bond.*

2. Features of the Peptide Bond

- *It is a covalent bond that stabilizes protein structure.*
- *It is coplanar, rigid, and planar. The 4 atoms (C, O, N, H) lie in a single plane.*
- *The Oxygen of the carbonyl and the Hydrogen of the amide nitrogen are usually trans to each other.*

- *Resonance: It exhibits a partial double-bond character due to resonance (delocalization of electrons between Carbonyl O and Amide N). This makes the bond strong and limits rotation around the C-N bond.*

3. The Polypeptide Chain

A series of amino acids joined together. Each unit is called an amino acid residue.

- *Polarity: The chain has directionality.*
  - *N-terminal: The left end, starts with the free* **α-amino** *group.*
  - *C-terminal: The right end, finishes with the free* **α-carboxyl** *group.*
- *Backbone vs. Side Chain: The repeating N-C-C part is the main backbone (which has hydrogen bonding potential), while the variable* **R-groups** *form the side chains.*
- *Naming: Read from N to C terminal. Suffixes change to -yl except for the C-terminal (e.g., Glutamyl-cysteinyl-glycine).*

4. Cis vs. Trans Configuration

Refers to the position of the two C_α *atoms relative to the peptide bond.*

- *Trans configuration:* C_α *atoms are on opposite sides. Generally Favored! It experiences minimum steric hindrance.*
- *Cis configuration:* C_α *atoms are on the same side. Extremely unfavorable due to steric clashes between bulky R-groups.*

5. Rotation & The Ramachandran Plot

While the peptide bond (C-N) is rigid, rotation is possible around the two single bonds adjacent to the alpha-carbon:

- ***Φ (Phi):*** *Angle of rotation between* $N - C_\alpha$
- ***Ψ (Psi):*** *Angle of rotation between* $C_\alpha - C$

The Ramachandran Plot

A graphical representation (X-axis: **Φ**, Y-axis: **Ψ**, both from -180° to +180°) showing allowed and forbidden conformations of amino acid residues.

- Purpose: Identifies energetically favorable regions, regions restricted by steric clashes, and validates 3D protein structures (X-ray/NMR).
- Darkest regions = most favorable combinations (e.g., **α**-helix, **β**-sheets).
- Exceptions:
  - Glycine: Smallest (H side chain), highly flexible, broader allowed regions.
  - Proline: Rigid ring structure, heavily restricted **Φ / Ψ** angles, limited regions on the plot.

6. Disulphide Bonds

Formed by the oxidation of sulphydryl (**-SH**) groups between two Cysteine residues. The resulting linked molecule is called Cystine.

Function: Highly important covalent cross-links that stabilize the tertiary structure of proteins (e.g., connecting the A and B chains in Insulin).

7. Biological Roles of Peptides

Category	Examples & Functions
Hormones / Pheromones	Insulin (sugar uptake), Oxytocin (childbirth)
Neuropeptides	Substance P (pain mediator)
Antibiotics	Polymyxin B (Gram -), Bacitracin (Gram +)
Toxins / Protection	Amanitin (mushrooms), Chlorotoxin (scorpions)
Others	Glutathione, Angiotensins, Bradykinin